Archaeological Excavation at Barabti Fort, Cuttack

2023-2024

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Introduction

The Barabati fort (20º 29'04" N, 85º 51'54" E, MSL 53m) is situated within the city of Cuttack, Odisha and at a distance of about 28 km north-east of Bhubaneswar the state capital. The fort was declared as a protected monument under the Archaeological Survey of India in 1915. The fort was strategically built at the acme of an island formed by river Mahanadi on the north and its distributary Kathajodi on the south. The fort spans an area of 102 acres, roughly rectangular in plan with a moat around locally known as *Gadakhai*. The *Gadakhai* (moat) of the fort was connected with Mahanadi through a breach in the levees on the right bank of river Mahanadi. The fort adorned with an enormous square structure at the centre and a carved gateway in the eastern side. The fort is an example of *Jala durga* (*jala*= water, *durga*= fort) of medieval Odisha. It is said that the name Barabati originated from the term *Barahati*, where Bara stands for twelve and Hati (a local unit of measurement for land).

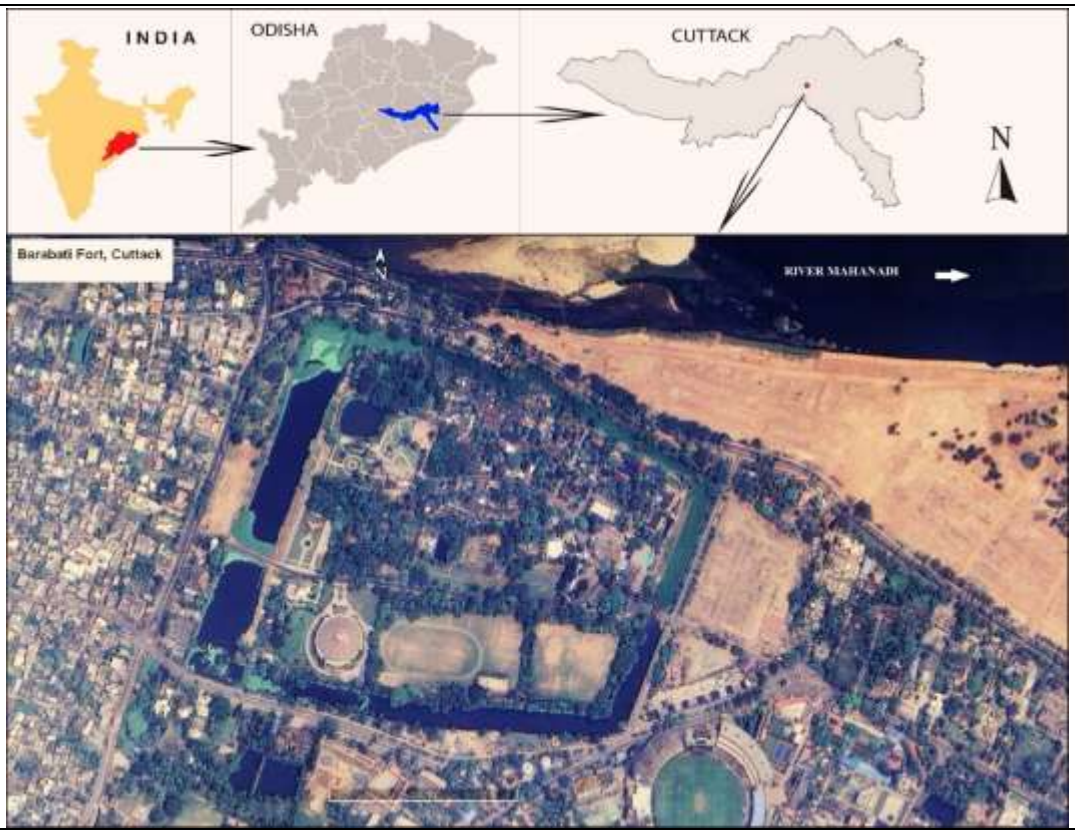


Fig 1. Location map of Barabati Fort, Cuttack, Odisha

The name Kataka / Cuttack in Odiya denotes a fortified town or city, generally meant to be a military establishment. In Sanskrit, the word KATAKA means a military camp, fort, or government seat protected by an army. The king Nrupa Keshari of Keshari / Somavamsi dynasty established a city in this place in 989 CE. The fort reached its glory during Anangabhima Dev III of the Ganga dynasty in 1229 CE, who shifted his capital from Choudwar Kataka to Abhinab Varanasi Kataka, i.e., the present Barabati fort at Cuttack. In the Madalapanji (Puri chronicle), it has been mentioned that the fort was built by Anangabhimdeva III (1211 CE -1238 CE). This tradition is supported by Nagari Copper Plate inscription of Anangabhimadeva III (Saka year 1151- 1152). In this inscription, Cuttack is referred to as Abhinava Varanasi Kataka. It also refers to the construction of a Purushottama temple in his capital Abhinava Varanasi Kataka. He issued land grants to a Brahmana named Devadhara in Kodinda Visaya for the maintenance of the temple. The Puri copper plate of Narasimha IV also mentions the construction of the Purushottama temple. It is mentioned that before giving land grants, the king purified himself by worshipping 'Purushottam -Jaganatha' in *Bhitara Navara* of the Kataka of Varnasi. (*Bhitara* = Inner, Navara here indicate city/palace). The two copper plates testify that the palace, possibly Barabati fort, had two parts during Ganaga rule; the inner part and the outer part where this Purushottam temple is believed to be constructed inside the palace. (B. C. Chhabra and D. C. Sircar, 1949-50. Nagari Plates of Anangabhima III, Saka 1151 and 1152, Epigraphia Indica (EI) Vol. XXVIII, pp-235-258). In Tarikh-i-firoshahi, it is stated that the temple of Purushottam was demolished and the idol was carried off to Delhi during the invasion of Firozhah Tughlaq in 1361 CE (H. M. Elliot 1871, History of India Vol III).



Fig. 2 Aerial view of Barabati Fort

Previous Archaeological Excavations

The first excavation at the site started on 1st December 1989 by B.K Sinha and continued by him till 1993 (1989 to 1993), and subsequently excavation had been conducted by K V. Rao (1994 -1997) followed by P.K. Trivedi (2003-04 & 2006-07) of Archaeological Survey of India.

Excavation of 1989-90

The aim of the excavation in 1989-90 was to establish a cultural sequence at the site in view of the existing tradition that the city of Cuttack was founded in 989 CE, and to conduct excavation at the mound to expose the remains of various cultural periods. During the excavations, a square structure built up of dressed khondalite stones was exposed and another structure was found having evidence of lime flooring. The building material of these structures is mostly reused architectural fragments and sculptural panels of temples. The important objects retrieved in the spade work are of broken sculptures and architectural fragments, pieces of Chinese porcelain, iron nails, empty brass cartridges, lead bullets, stone cannon balls, stem and base of a glass goblet and some earthen lamps etc. (IAR 1989-90, pp 74-77).

Excavation of 1990-91

The excavation at Barabati fort was started with the objective to know the cultural sequence, to expose the complete plan of the structure (palace), and to fully expose the partially exposed citadel wall on the southern and eastern side of the structure. The excavator identified three cultural periods during this excavation, i.e., early medieval period (13th -14th century), medieval period (15th -16th century), late medieval period (17th-19th century). The elements of temple along with sculptural members, have been documented along with a pillar hall in the excavation. During the excavation, it has been noticed that the entire palace was filled with five-meter-thick sand and debris. The important artifacts retrieved from the excavation are of iron arrow heads, sculptures of musicians, erotic couples, maidens, terracotta beads and a silver coin of Shahjahan, the Mughal emperor, which was minted at Surat (IAR 1990-91, pp 53-54).

Excavation of 1991-92

In continuation of the previous field season's excavation in 1991-92, the excavator focused on tracing out the foundation of the temple. During excavation, the mouldings of the temple have been traced out along with beautifully decorated stone slab and floor and a pilaster depicting beautiful sculptures representing erotic couples, musician female head, decorative architectural members, and *chauri* bearers. These sculptures are stylistically dated to the 13th century CE. The excavation in the area east of the citadel wall brought to light a gateway complex and the exterior of the gateway was decorated with mouldings. Excavations in the north of the palace showed a plastered wall (length 13 m) east-west forming the northern corridor of the palace. It had a thickness of 2 m and its outer side also contains lime plaster (IAR 1991-92, pp 83-84).

Excavation of 1992-93

The aim of the excavation was to uncover the remains of the temple partially discovered in previous year, to trace the plastered wall of the northern corridor of the palace on the eastern side and its working level further east, to expose the eastern and western sides on the northern flank of the citadel wall and the Darbar hall down to its earliest working level, and to locate the south-western corner of the citadel wall and the wall on the western side apart from the stone structures lying in the south of the citadel wall.

The platform of the temple, excavated in the previous year, was found extending on the western as well as on the southern side. On the western side the

platform was found extending below the Durbar hall of the palace, while on the southern side it was traced across the southern corridor, beyond pillars. The platform was available to a length of 30m east-west and about 22m north-south. The plastered wall, forming the northern corridor of the palace, traced up to a length of 13m was further traced on the eastern side. Two structural phases have been noticed in the operation area. The excavation uncovered remains of more pillars made out of laterite blocks as well as evidence of laterite structures. Evidence of a drainage system was also noticed in the citadel area. Noteworthy findings from the citadel area were a piece of a small bowl of Celadon ware and some bowls and flat dishes of Chinese porcelain. Besides, a good number of damaged stone sculptures were also recovered during excavations which represent inverted lotus pedestals, friezes showing procession of elephants and male and female figurines, fragments of squatting lions, architectural fragments fauna and floral motifs; iron arrow-head, terracotta animal and human figures etc. (IAR 1992-93, pp 78-81).

Excavation of 1993-94

The excavation started by B.K Sinha from 1989-93 was later on initiated by K. V Rao from 1993 to 1997. The objectives of the excavation in the said years to fix the structural phases at the site. There are four structural phases noticed during the excavation starting from the early 12th century up to 16th century CE. The structural activity at the site could be divided into four periods. On the basis of architectural style Period I, structures can be dated to 12th century CE. The structures of Period II were dismantled and the material was used for structures of Period III. But the plastered wall was not completely destroyed as it was probably incorporated in the scheme of structures of Period III. The construction of a *Nabatala prasada* by Raja Mukunda Deva dated around 1560-68 CE may be the III phase of structural activity at the site as per the director of the excavation, which was also corroborated by the accounts of Abul Fazal (Ain-i-Akbari).

The outer wall of tower provided with steps in the shape of a ramp on the northern side belongs to period IV architecture. The tower portion was enlarged subsequently. This was most probably built during the Muslim period after the death of Raja Mukunda Deva in 1568 CE. In the medieval period these were called *burz* and primarily meant for keeping watch for safeguarding the fort. The excavation revealed architectural members, sculptural remains of divinity and semi-divinity and Coins of copper and potin with Islamic characters on the obverse and reverse (IAR 1993-94, pp 84-88).

Excavation of 1994-95

The excavation in the field season of 1994-95 revealed rest of the retaining wall exposed in the south-western side below the tower which may be raised to protect structures from flood damage. The exposed wall (7.75 m) was oriented north-south and had an outer face with steps and an interior face that was not plumb line. The wall (1.65 m) was constructed of dressed laterite blocks (80 × 40 × 25cm). It had a step-like appearance on the outer face. The ceramic assemblage yielded from the excavation includes grey ware with medium fabric, dull and red slipped ware and Chinese porcelain with painted designs such as houses, trees, birds, and butterflies. Not a single specimen contained Chinese characters. The shapes were mainly *handis*, vases, basins, small vessels with carination, spouted vessels, and vases with corrugated neck, *surahis* and even receptacles of *hookahs* (hubble-bubble). Apart from this ceramic assemblage, wheel-made and lathe-turned soapstone vessels have also been found. The important antiquities of this field season are a copper coin, terracotta figurine, and a stone-lamp with a lotus motif (IAR 1994-95, pp 59-60).

Excavation of 1995-96

K. Veerabhadra Rao conducted excavations on the north-eastern corner of the mound, revealing the eastern end of the northern retaining wall of the citadel from Period III (1560-68 CE). The wall was built of dressed laterite blocks with mud-mortar, measuring 85× 8 × 22 cm on average. The exposed portion had eleven courses and was badly damaged due to robbing of stones. The remaining floor above the mound was exposed and documented. In the southern side of the Mazhar, a drain was found at a depth of 1.12 m, built of dressed khondalite blocks. A structure resembling the foundation of a wall was found in three courses, laid alternately along the length and across it with khondalite blocks. The grey ware and red ware were mainly retrieved from the southern side. A copper coin bearing the legend '*Shah Alam Badshah Sanah Julus'*, hand-made terracotta figurines of animals, a female sculpture in high relief decorated with ear-studs and a waist band have been revealed from the excavation (IAR 1995-96, pp 61-64).

Excavation of 1996-97

The aim of this year excavation was to focus on exposing the structures surrounding the palace and tracing the relationship between the heavy foundation columns and the retaining wall at the north-eastern corner. Due to encroachment, excavation was restricted to the western side, with the southern edge adjoining the road leading to the Jawaharlal Nehru Indoor Stadium. The western retaining wall of the

citadel was covered under 9 meters of excavated earth, and the area on the southern side of the Palace area was covered under almost 3 meters of dump.

Ceramic assemblage mainly represented by Chinese porcelain- a shallow plate with ring base painted inside blue colour depicting huts, trees, boats etc. Other than this pottery such as grey ware and dull red ware with slip of medium fabric and red slipped ware of medium to fine fabric were also recovered. The major shapes are vases, shallow dishes, bowls and *surahis*. The excavation revealed architectural fragments like pillar bases, brackets, door-jambs, stone-channels, and other minor architectural fragments. Most of the architectural fragments are carved out of *khondalite* stones, while robust columns, pillar-buses, and door-jambs are carved out of schist with a considerable degree of polish. Five pieces of columns were found measuring 112 cm to 144 cm in height and having a section of roughly 80 cm. At the north-eastern corner of the Palace area, heavy foundation columns were found built on existing floor-level after the construction of a retaining wall. The other finds from the excavation includes terracotta figurines, a badly damaged female sculpture, and fragmentary Chinese porcelain fragments (IAR 1996-97, pp 79-82).

Excavation of 2003-2004

The excavation of 2003-04 was resumed by P.K. Trivedi with an aim to expose the structures and co-relate with the previously excavated one. The excavation revealed 3.60m cultural deposits at the site, which have been divided by the excavator into two periods viz. Period I and Period II. Period I has two cultural phases in which Phase I were encountered with two parallel structures made out of laterite blocks and oriented towards north-south. Structure I measures 1.90 × 1.00m and 1.56 × 1.00m whereas the structure II is T-shaped measuring 2.5m wide. In the phase II another stone-built structure (Str. III) yielded on the northwest corner probably representing a pavilion. It is a plain course of laterite masonry retaining the evidence of lime plaster. The floor of the structure was paved with sandstone, laterite and blue greenish schist stone. Stylistically it is dated 16th century CE.

Period II is dated to 17th -18th century CE in which remains of a male elephant tusk and large burial pit were found. Apart from this, eighteen massive pillar bases raised over the fillings were unearthed and these are mostly made of laterite. Few architectural fragments of sandstone have also been occasionally displayed. A ring well is exposed of 7- 10 rings and inscribed in Oriya numerals. It can be dated to 18th century CE.

The pottery assemblage includes red ware, black slipped are, grey ware and red slipped ware which are made of well levigated clay, wheel thrown and well- fired. Important shapes comprise of vase, basin, jar, trough, bowl, dish, *handi* and *surahi* etc.

Stone figures of a dancing male, female head, standing couple, circular spheroid, head portion of a sculpture, three animal figurines and lamp of black chlorite schist etc. are important antiquities unearthed during the course of excavation (IAR 2003-04).

Excavation of 2006-2007

Excavation was resumed at north-eastern part of Barabati under the supervision of P. K. Trivedi. It emphasized to uncover the burial structures and link with the earlier ones. About 4.30m thick cultural deposit was dug up which can be assigned to 14th to 17th century CE. Two structural phases have been exposed. In Phase I, a 3.9m wide and 12.25m long wall running east-west below the eastern arm of citadel was found in the continuation of the previously exposed corridor. It has eleven courses. Extension of already exposed laterite wall (5.50 long and 4m wide) found during the spade work which is belongs to the Phase II. Structures of both phases are made using laterite, *khondalite*, coarse sandstone and lime mortar plaster.

Important pottery shapes include spouted vessels, storage jars, lamps, knobbed lids, pot stands, dishes, miniature pots and pieces of *hukkah*. Female figurines, erotic sculptures, bearded figures, bust parts of male and different architectural members, and various types of iron objects are also recovered from the excavation (IAR 2006-07, pp. 55-65).

Present Archaeological Excavation (2023-24)

The excavation at Barabati Fort was conducted at the site during the 2023-24 field season (from October 2023 to May 2024). Vegetation and debris clearance were done before laying out trenches in three locations. Trench numbering was based on the previous year's excavation grids. The grid methodology was adopted for the excavation by laying out 10x10m trenches with four quadrants.

Aims and Objectives

The excavation at the site resumed with an aim to find out the material culture which can linkage between Odisha and South-East Asian countries and beyond; to find out tangible evidence to prove the local tradition and folklore about the rich maritime history of Odisha; to trace out the material culture before 12th -13th century

CE; to preserve and conserve the exposed suitable structure for public and researchers.

The Excavation

Three locations have been chosen for excavation, named as Area I, II, and III, as per the availability of the space. Area I is in the central part of the site, near the pillared hall exposed during the excavation in 1991. Area II is located on the southern side of the Shahi-Qila mosque. Area III, south-east of the recently developed Museum of Justice. The spade work was conducted based on the surface study, available space, observing the already exposed structures and existing buildings, and by studying the elevation of the site from the contour lines.



Fig. 3. Bird eye view of the excavated areas I, II, III

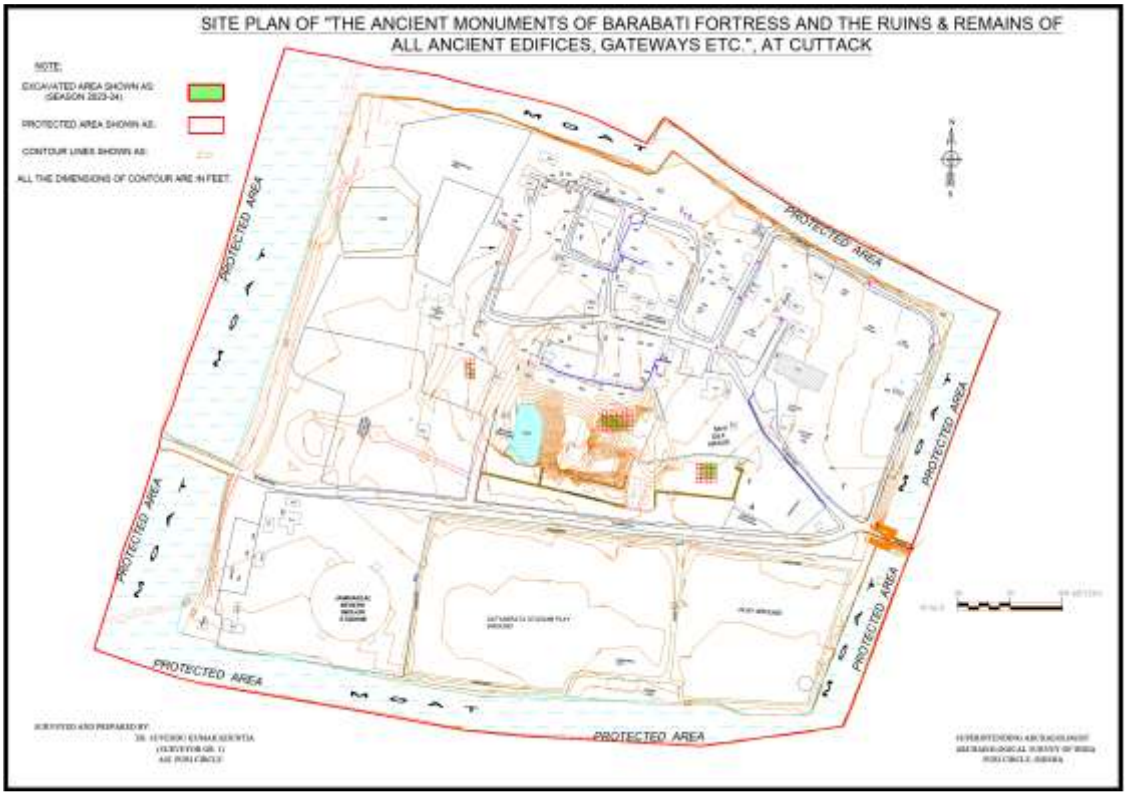


Fig. 4. Site plan and contour map of Barabati Fort

3. Stratigraphy

The excavation at the index trench in Area II revealed seven cultural layers without any hiatus, starting from the top to bottom, with a 4.25m cultural deposit. After the removal of 15 to 30 cm of humus layer, which is light brownish grey colour (10YR-6/2), loose in nature, fibrous and contains stone nodules, potsherds and patches of *murram*, a proper cultural stratigraphic deposit has been noticed in the site.

Layer 1: Brown in colour (7.5YR-5/3), semi compact, coarse in nature and mixed with stone chips, laterite and sandstone blocks, brick bats, ceramic evidences of red ware, grey ware, few sherds of porcelain. The maximum thickness of this layer is about 60cm.

Layer 2, the layer is semi-compact to loose, medium to coarse grain, reddish brown in colour (5YR-5/4), mixed with laterite nodules, stone blocks and flakes, architectural members both of sandstone and laterite, brickbats, and ceramics of grey ware, red ware, and sherds of porcelain, celadon. The artefacts in terms of terracotta figurines, iron nails, copper objects, cowries, glass bangles, etc., are observed; the maximum thickness of the cultural deposit is 90cm.

Layer 3, has a maximum 92 cm**,** light reddish brown (5YR-6/3) colour, semi-compact to loose in nature. Most of the structural as well as pit activities have been observed in this layer. Iron and copper objects are found in good numbers, along with traces of lime plaster and terracotta tiles. The percentage of porcelain, celadon, is higher in this layer in comparison to the upper layer. One dump has been noticed within the layer along with patches of burning activities.

Layer 4, reddish brown in colour (2.5YR-5/4), semi-compact to compact, coarse in nature and mixed with laterite nodules, sherds of ring well and evidence of rammed floor of *murram*, ceramic evidence includes grey ware, red ware and a few sherds of porcelain, stone ware noticed at this layer. The structure, having 50cm of thickness noticed in layer (3), is of laterite stone.

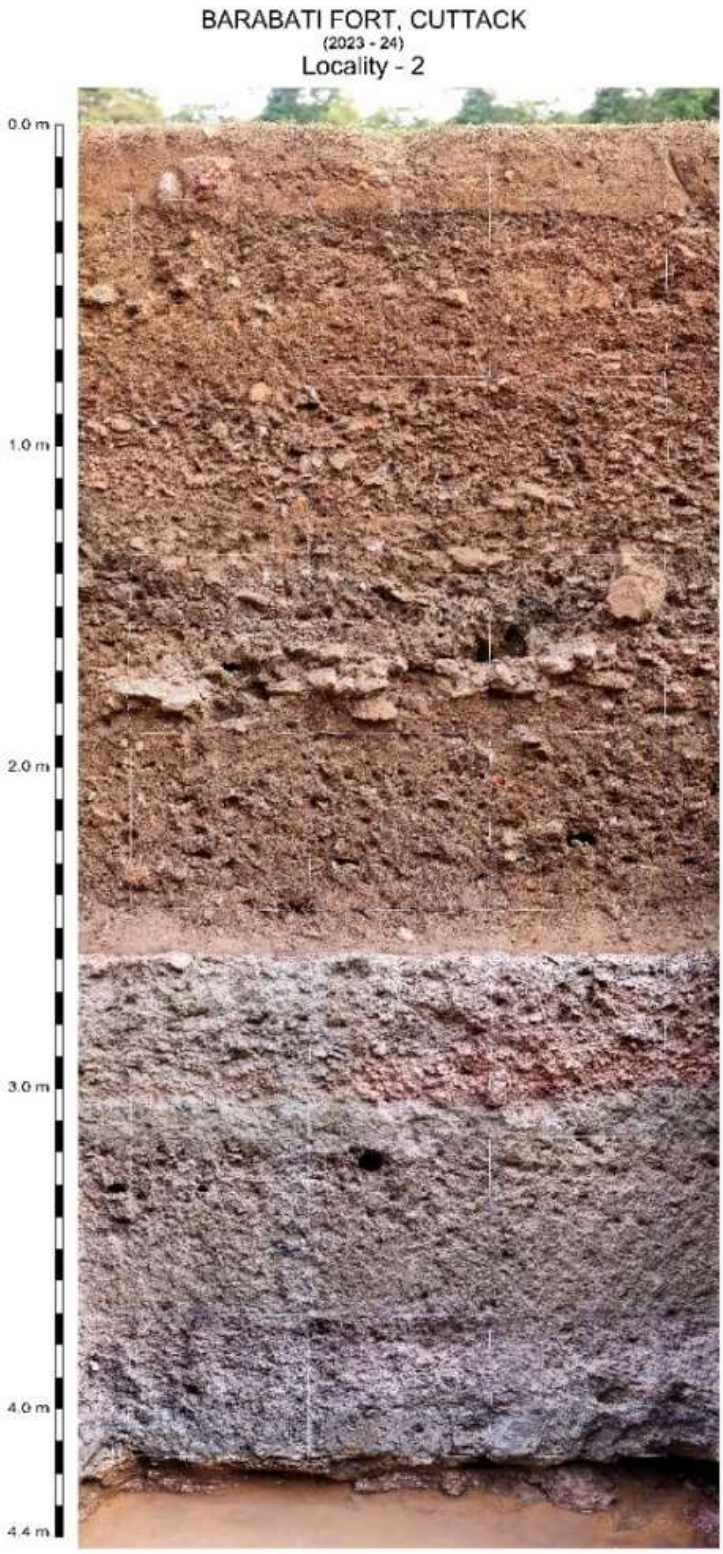


Fig. 5. General view of section facing north

Layer 5, is brown in colour (10YR-4/3)**,** semi-compact, clayish in nature, and presently under the ground water level, and has fine nodules and is mixed with laterite

blocks and ceramics such as grey ware, red ware, and black ware. Its maximum thickness is 15cm.

Layer 6, Very dark greyish brown in colour (2.5Y-3/2), compact, sticky, and clayish in nature. Ceramic evidence observed in Layer 5 is continuing, but with finer fabric. Apart from this, a few sherds of white ceramic with a crackled surface have been retrieved. A laterite block-built structure has been noticed in this layer, which is supposed to be the earliest structural evidence of the site dated before the 9th cent CE. The maximum thickness of the cultural deposit is 60cm.

Layer 7, colour is light yellowish brown (2.5Y-6/4), sandy in nature, mixed with small sherds of grey, red, and black ware. In this layer, a flood deposit is noticed. Its maximum thickness is 45cm.

Within the sub-soil water, the excavation has been carried out up to a depth of 5.60m. The nature of the layers below the 7th layer is of light coloured sandy in nature, followed by a yellowish granular sand deposit. The auger method done below 4.25m has revealed 13cm of white sand (4.25m - 4.38m), 30 cm of yellowish sand (4.38m-4.68m), and below the level fine fine-grained sand was observed. No trace of human occupancy below this level was discovered during the current excavation at Barabati Fort. To know the stratigraphy of the fort in 1994, the Orissa Water Supply and Sewerage Board also dug a bore well near Jawaharlal Nehru Indoor Stadium, on the southern side of the fort. According to the lithological chart, boulders and coarse sand were found at a depth of 170 feet. Above this, there are fine to medium sand for 60 feet, followed by 20 feet of fine sand, 40 feet of dark sticky clay, and 15 feet of top soil (K.V.B. Rao, 2000. Excavation at Barabati Fort, Cuttack, published in K. K. Basa and P. Mohanty (eds), *Archaeology of Orissa*, pp. 661- 680).

Based on ceramic and structural remains the whole cultural deposit can be assigned from the Early-medieval period (circa 7th cent CE) to early 20th century CE, covering the political history from Bhaumakaras, Somavamsis, Gangas, Suryavamsis, Gajapatis, Bhois, Afghans, Mughals, Marathas, till the colonial period.

Chronology of the Site:

On the basis of material culture in the form of ceramic assemblages, sculptural and architectural remains, and historical facts, based on epigraphical and literary sources, the history and chronology of Barabati can be classified into six cultural periods without any hiatus in the chronology. It is one of the sites in Odishan context that retains the history for more than thousands of years from the beginning of the 7th century CE to the present day. It beholds a strong, tangible maritime heritage and

folklore tradition of the state, the architectural remains, broken sculptures exhibit the struggle of Odias to retain the identity of Barabati from the beginning. The strategic location of Barabati carries forward its habitation from an emerging rural settlement to a fortified capital city and a major trade centre of the medieval period. The following time frame chronology can be assigned to the entire cultural phase of the site:

Period-I: 7th to 10th cent. CE (Bhaumakaras -Somavamsis Period),

Period-II: 11th to 13th cent. CE (Ganga Period)

Period-III: 14th century CE to 16th cent. CE (Gajapati -Bhoi and Afghans)

Period-IV: 1592 to 1751 (Under the Mughals)

Period V: 1751-1803 (Under the Marathas)

Period- VI: 1803 to 20th century (Under the Colonial rule to the Recent past)

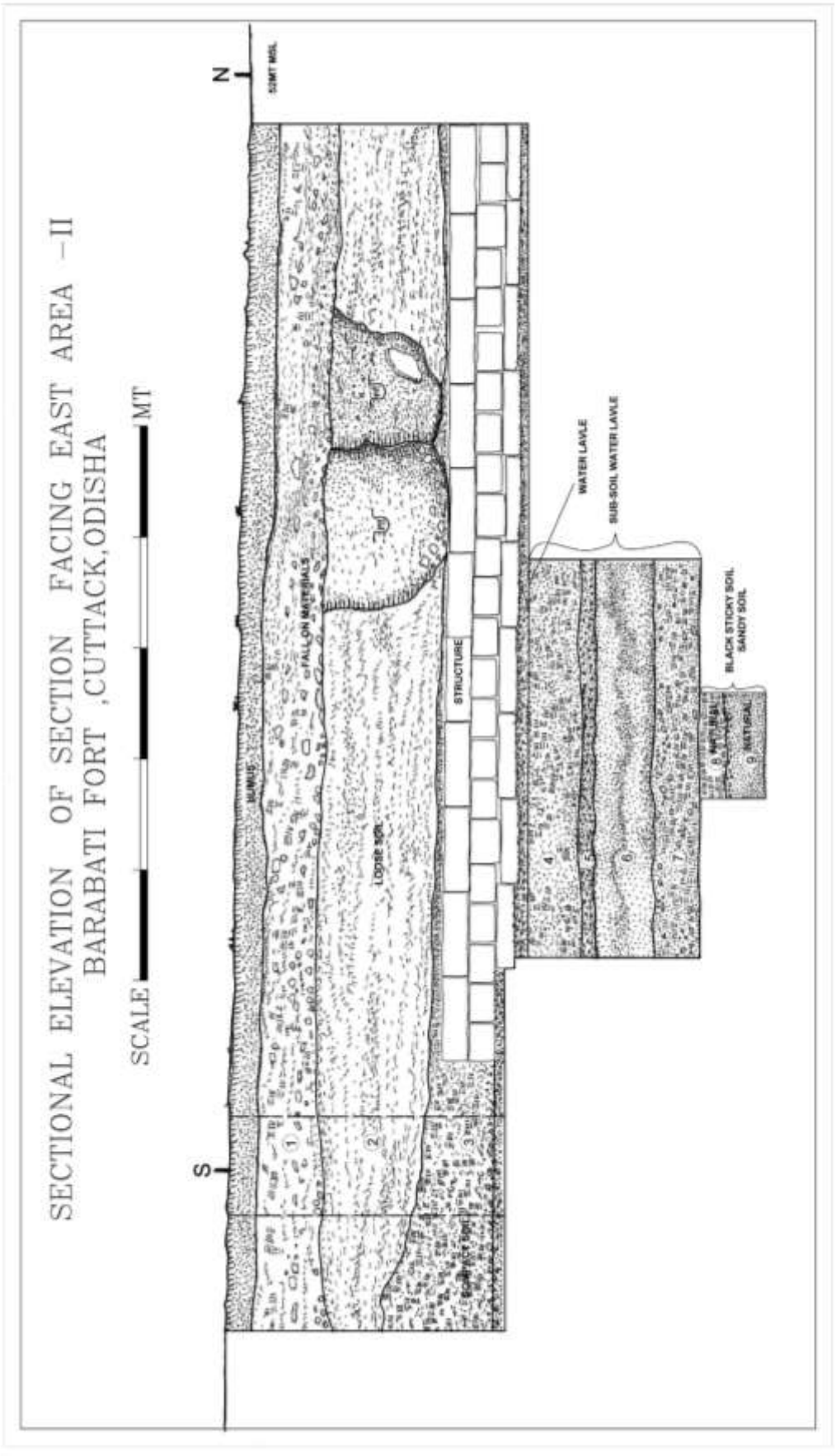


Fig. 6. Drawing of the section facing east

4. Features (Structures)

The spade work exposed structures and structural remains of both secular and religious in nature. The dressed stone paved pathway in Area I, stone-built walls and a complete plan of a laterite-built structure were noticed during excavation. The roughly rectangular structure is made up of laterite blocks measuring each wall 10.20m in north x 10.10m in south x 10m in East x 9.80m in west, having a width of 90cm to 95 cm and the existing height is 62cm. Three courses of laterite blocks were placed one after the other in a stretcher pattern with dry masonry. There is an entrance facing towards the south having a 1.65m opening. The partially exposed stone-built pathway made of sandstone blocks unearthed in the site shows a meticulous pattern and design of the medieval period, oriented from east to west having 24.24m L x 1.47m width and 15 cm height constructed over a prepared laterite block platform. Another laterite block built structure was encountered in the Northeast corner of the index trench during the excavation at the depth of 3.85m. The measurements of the exposed laterite block are 61cm in length and 22cm in thickness. Besides structural remains of secular in nature many religious edifices were encountered during the excavation such as lintel, decorative architectural members, sculptural fragments etc.

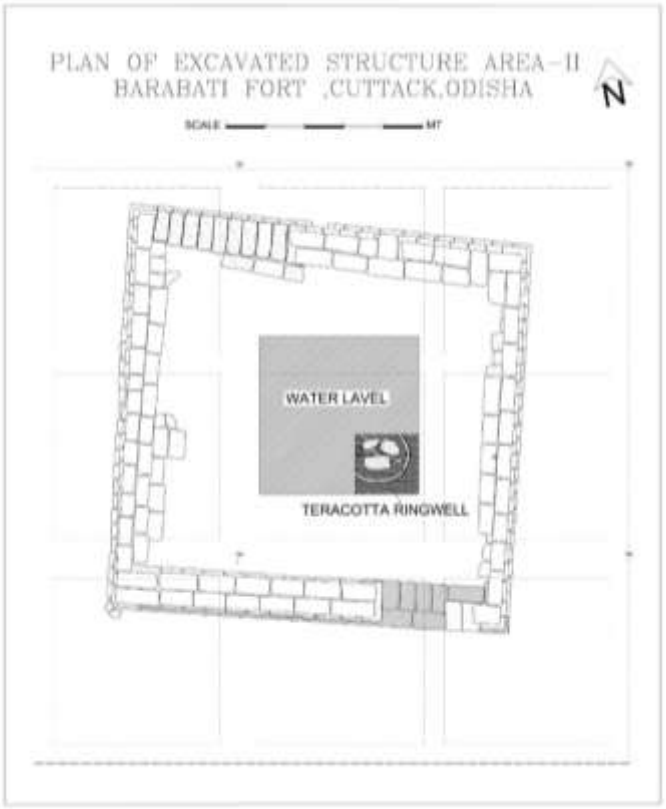


Fig.7. Plan of the excavated structure in area II

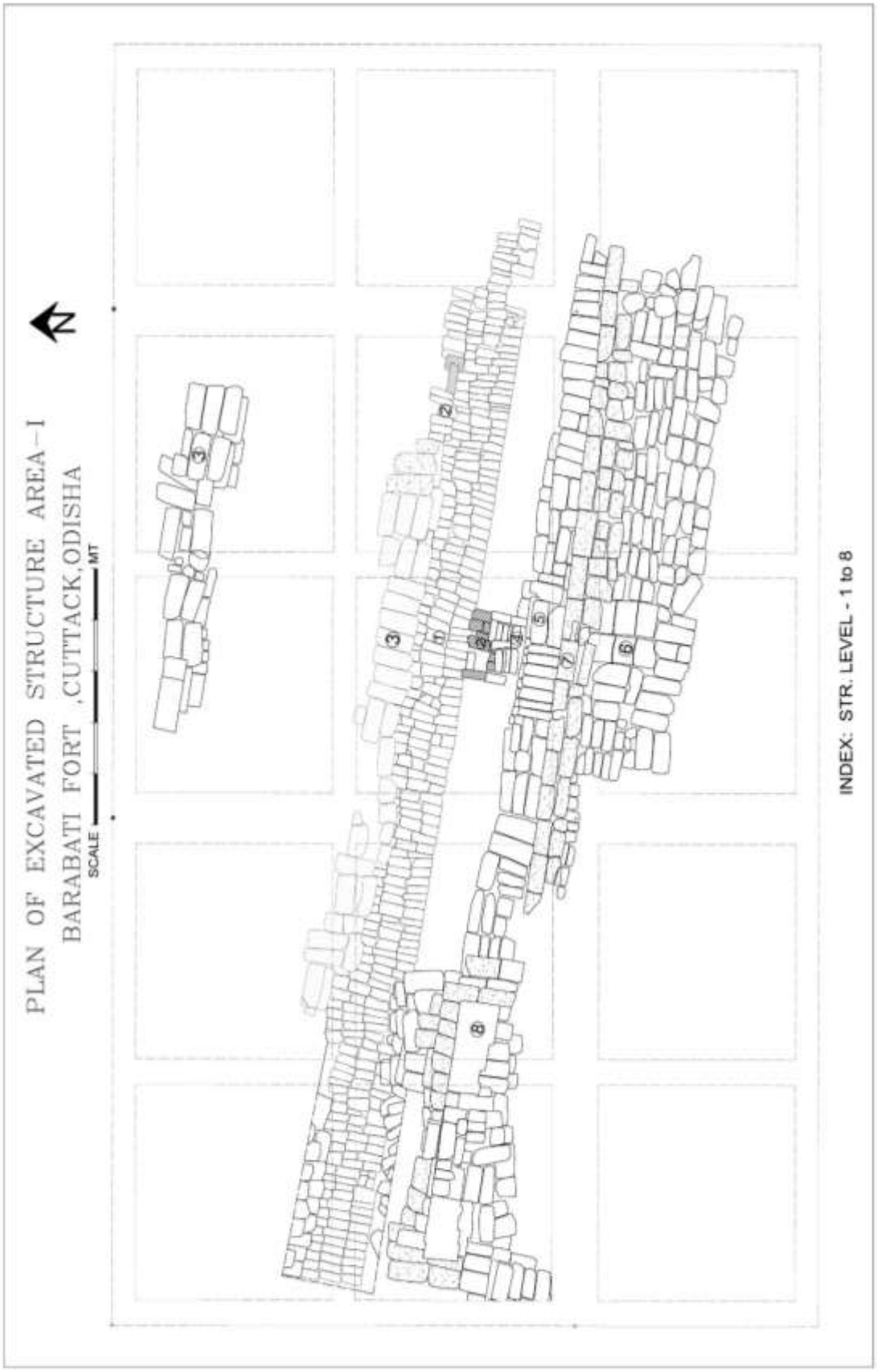


Fig.8. Plan of the excavated structure in area I

5. Findings

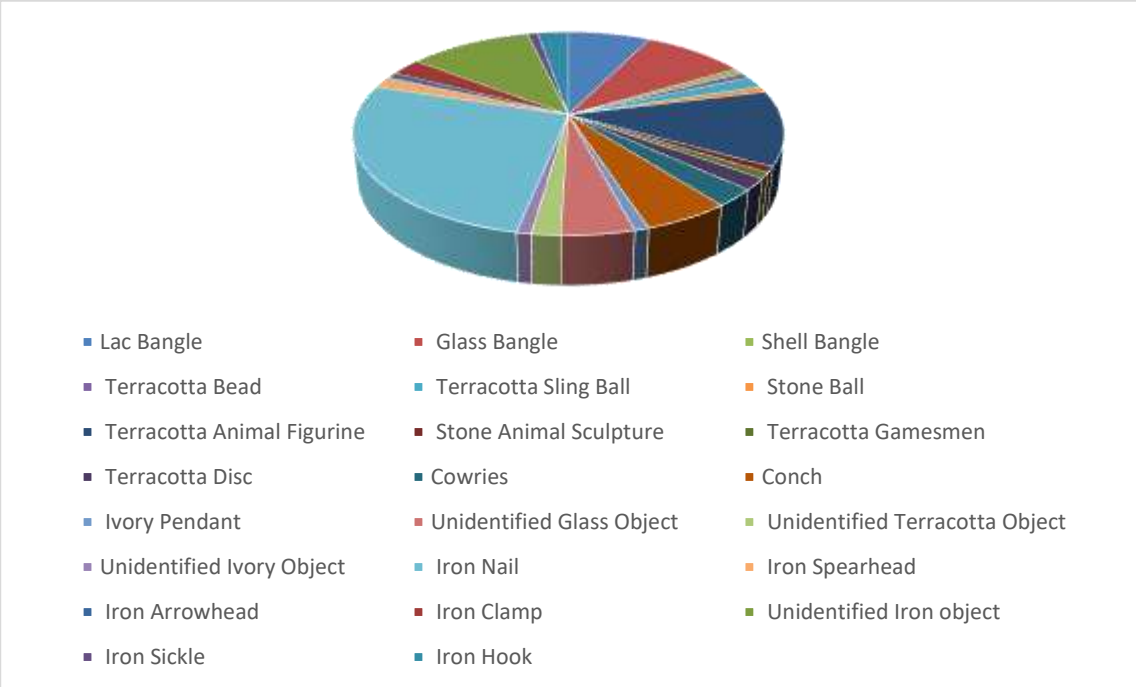
A. Antiquities

During the excavation terracotta, stone, metal and shell objects were yielded from the site. Iron and Copper objects, like elongated nails, hooks, arrowheads, ear studs, copper strings are retrieved. The art and craft industries were at their peak in medieval Odisha. The city of Cuttack is well known for its filigree works. The excavation in area II has a lot of evidence related to craft industries. Within a rectangular house structure, a lots of tiny copper objects, burning activities, gold-plated ornaments and different varieties of iron objects starting from micro iron bits to big size nails are documented. Another important feature of this house is that of a terracotta ring well within the house complex, which may be used for industrial activities. Most of the artifacts retrieved from the sites are of domestic as well as warfare in nature.

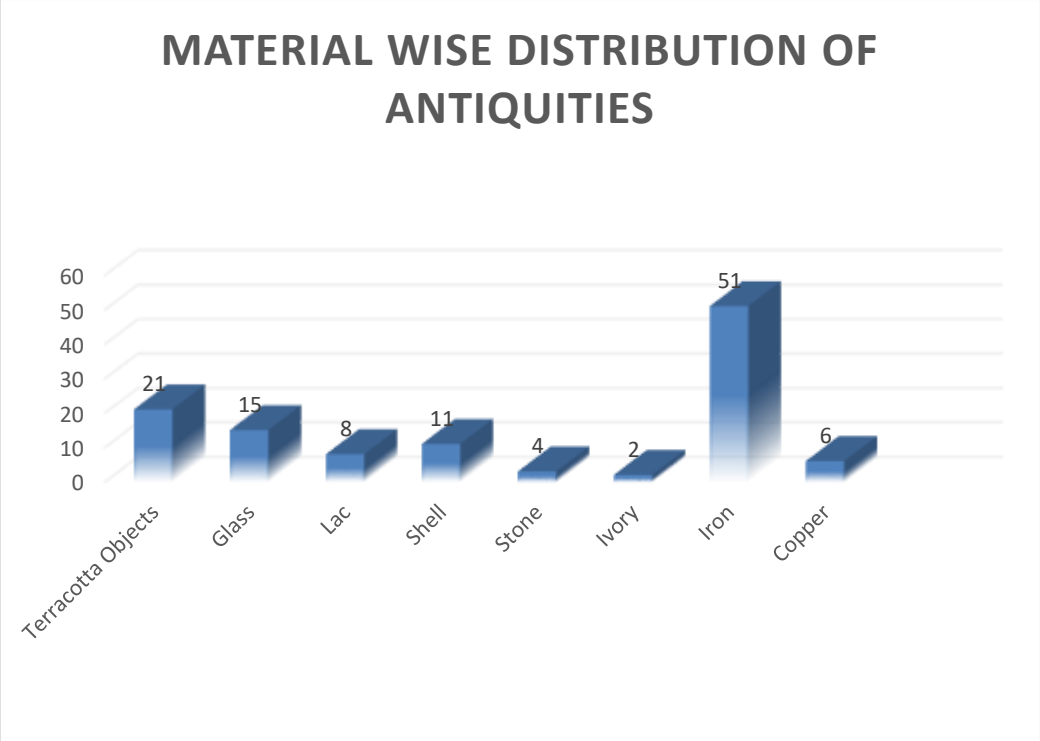
The list of the objects given below based on material:

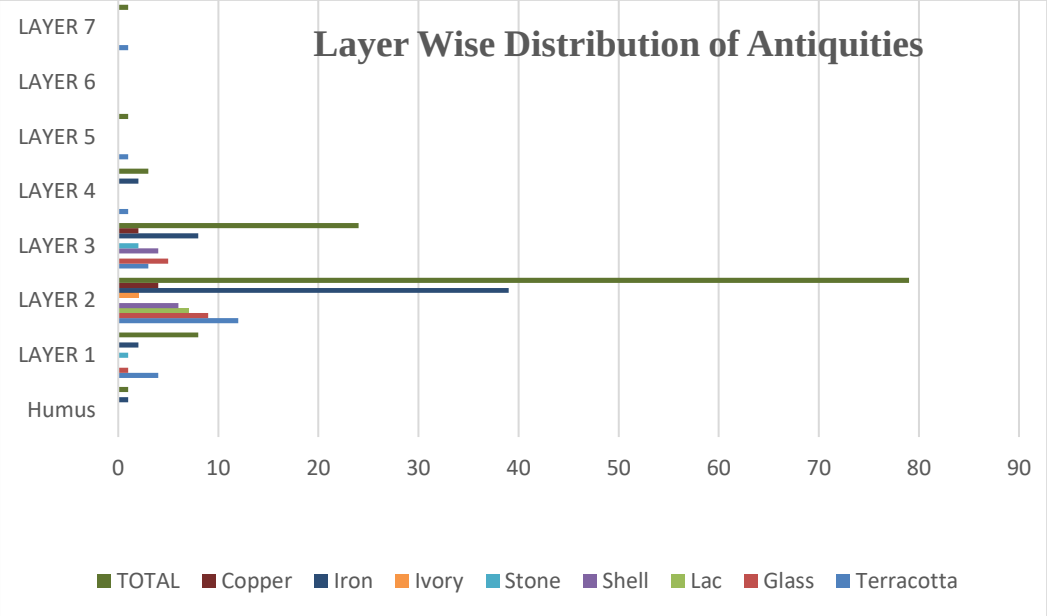
Sl. No.	OBJECT	MATERIAL	TOTAL NUMBER
1.	Bangle	Lac	8
2.	Bangle	Glass	10
3.	Bangle	Ivory	1
4.	Bead	Terracotta	1
5.	Sling Ball	Terracotta	2
6.	Ball	Stone	1
7.	Animal Figurine	Terracotta	13
8.	Animal Sculpture	Stone	2
9.	Gamesmen	Terracotta	1
10.	Disc	Terracotta	2
11.	Cowries	Shell	3
12.	Conch	Shell	6
13.	Pendant	Ivory	1
14.	Unidentified	Glass	5
15.	Unidentified	Terracotta	2
16.	Unidentified	Ivory	1
17.	Nail	Iron	29
18.	Spearhead	Iron	2
20.	Arrowhead	Iron	1

Sl. No.	OBJECT	MATERIAL	TOTAL NUMBER
21.	Clamp	Iron	3
22.	Unidentified	Iron	12
23.	Sickle	Iron	1
24.	Hook	Iron	3



Sl. NO.		MATERIAL	TOTAL NUMBER
1.		Terracotta	21
2.		Glass	15
3.		Lac	8
4.		Shell	11
5.		Stone	4
6.		Ivory	2
7.		Iron	51
8.		Copper	6
Total			118





Iron Objects:

Knowledge of iron tool manufacturing was a common phenomenon in early Odisha and useful metal for the villagers and urban people. One of the earliest copper plate grant known as Kondedda Grant of Dharamaraja (7th Century CE Sailodbhava family refers to manufacture of different weapons *(sastras*) to vanquish the enemies, which might have made of iron. R.L. Mitra, (Antiquities of Orissa Vol. I Pp. 123, 1875)

referred in his work that short swords or dagger are most probably of great favorite in Odisha. The warrior's sculptures found carved on temple walls holding daggers and short swords shows the manufacturing of such weapons in iron was prevailed in the medieval period. The presence of different nature of iron objects i.e. nail, chisel, arrowhead, single tang arrowhead, spearhead, hook, clamp, sickle etc. from the excavation shows advance metallurgy in mediaeval Odisha.



Fig.9. Iron spearhead



Fig.10. Iron nails



Fig.11. Iron nails from Area II, Barabati Fort

Copper Objects:

The present excavation in a limited area within the fort revealed few copper objects i.e., ear stud, copper wires, pendant, copper sheet etc. It shows flourishing craft specialization of the period as well as it is attested by the sculptures of early medieval Orissa displayed with variety of ornaments like earrings, ear flowers, necklace, waist ornaments, leg ornaments armlets. One of the early references about coppersmith found in the grants of Dandimahadevi of Bhaumakara family dated to 8th-9th century CE when the excavated site was in existence and part of Bhaumakara territory as attested by the remains of material culture.

Cowrie shell:

The excavation revealed lots of cowries starting from the bottom, the maximum number of cowries yielded from layer 3 of Area III which can be dated to 13th -14th century CE on the basis of other associated material culture and overseas ceramics. Generally, in the economic history of medieval Odisha, it has been observed that cowrie was a medium of exchange and has religious value. It is continued to be used in various transaction along with other medium of exchange as late as 18th century CE. The cowries formed the basic unit of monetary calculation until recently in the arithmetical tables found in Odiya, Bengali and other regional languages, known as *kadakiya*, *panakiya* etc. According to this calculation- 4 *kadas* or cowries = 1 *ganda*, 20 cowries= 5 *gandas* or 1 *bodi*, 80 cowries = 20 *gandas* or 4 *bodis*= 1 *pana*, 1280 cowries = 320 *gandas*= 64 *bodis*= 16 *pana*= 1 *kahana* or a silver rupee (S. Tripathy, 1986, Early and Medieval Coins and Currency System of Orissa).



Fig.12. Cowrie shell, Barabati Fort

Bangles:

Bangles have been recovered of glass, shell and lac from the excavation. These are either monochrome or bichrome and some specimens of polychrome. Most of the bangles found at the site are having maximum and minimum thicknesses are of 0.8cm and 0.2cm respectively. The diameter varies from 5.14cm to 2.1cm.

Conch:

The Conch shells (*Shankha= Turbinella pyrum*) are retrieved in Area II of the excavation, few of them are finished and semi-finished. *Shankha* is an auspicious object and used as a sacred symbol associated with many religious sects. It is one of the *Ayudha* (attribute) of Lord Vishnu/Purushottam, a Vaishnavite deity of Brahmanical pantheon. In archaeological context sometimes *shankha* motifs are found depicted in pottery as well as in sculptural art from the early historic period to late medieval period. The spade work at the site yielded *shankha* in layer 2 which can be dated to 12th -13th century CE. As per the literature and in the Nagari copper plate of Anangabhimadeva-

III, a temple dedicated to lord Purushottam was adorned at the site dated to 12th -13th century CE.



Fig.13. Conch shell, Barabati Fort

Terracotta objects:

Different kinds of terracotta objects such as animal figurines, sling balls, terracotta beads etc. have been found during the spade work. They were usually thrown out but these fragments have almost become permanent and survived as a record of past human activity in the habitats. They are found profusely scattered on the habitat and show distinct stratigraphic superposition and hence excellent evidence for the chronology and human activity. Most of the figurines at the site include that of zoomorphic in nature. Bull, elephant and horse with saddle are the most common figurines found in Barabati.



Fig.14. Miniature Pot



Fig.15. Terracotta objects

Stone Objects:

The excavation yielded a number of stone objects of religious and secular in nature, some are used in the household and few are used in warfare. A circular in cross section granite stone pestle was yielded from the excavation. It is broken from the centre may be used for grinding corn and spices. A few spherical stone balls were

retrieved from the excavation. These spherical balls were probably used as weapons for two-string bows and the bigger stones were hurled by hands or catapults of different sizes.

Sculptures:

One unidentified sandstone sculpture of an animal (humped bull ?) on a pedestal has been found. The head of the animal is missing and the back portion is visible. Another beautiful panel of elephant along with a lioness in chasing position has been marked. The animal figurines are exhibited in alto relief, smooth surface, mobile in nature may be a part of decorative panel of a temple of 12th -13th century CE.

A lintel made out of huge sand stone block of a temple has been documented during the excavation. In the center niche of the lintel housed goddess Parvati having Ganesh in her lap which is very unique and rare in the sculptural art of Kalingan order, none of the temples of mediaeval Odisha has such evidence. It shows that it might have influenced from southern school of art tradition borrowed by Gangas in 12th -13th century CE. Besides the lintel carved with *sakhas* (molding) decorated pattern with *nayika,* flower design, scroll work and at the top of the lintel, above the niche has been carved with *graha* panels.

B. Pottery

The excavation revealed pottery/ceramic assemblage includes both of indigenous and foreign / overseas variant. In indigenous variant the dominating wares are of red ware followed by grey ware, black ware, of dish, *handi*, basin, vase, miniature pot, spouted vase, bowl, trough, *surahi*, knobbed lid, footed pot and lamp etc. The pottery assemblage found is mainly wheel thrown, well fired and made of levigated clay available from in the alluvial of Mahanadi area the overseas variant pots generally of porcelain (plain and painted), celadon, Burmese ware, white ceramics and glazed ware. The important shapes are of bowl, cup and dish having different fabrics and texture.

Red Ware

The dominating variety in comparison to the other ware is red ware with variant of polished red ware, reserved red slip ware, red wash ware plain red ware is observed during the spadework. The major portion is pervaded with wash while slip is noticed in some examples. *Handi*, vase, shallow dish, basin, dish, lamp, deep and shallow bowl and miniature pots are the most common shapes retrieved from the digging. In the upper layers coarse to medium fabric is noticed whereas more fine fabric is seen in the

bottom layers. The thickness of the sherds ranges from 0.19cm to 1.89cm. Some sherds are found with decorations. The decorative designs include horizontal bands on neck, nail headed notches and grooving on shoulder.

Grey Ware

Grey ware is one among the leading variety followed by red ware found from the excavated area. Different shapes such as carinated *handi*, vase, dish, storage jars, lamp, various types of bowls and shallow dish are unearthed. Both polish and wash is observed in grey ware and slip is noticed in rare cases. All of them are wheel made and the thickness of the sherds varied from 0.3cm to 2.21cm. From the bottom cultural layers of the site, more refined and fine variant of grey ware retrieved. Designs such as horizontal bands, grooves are also observed on sherds.

Black Ware

The Black ware yielded throughout the deposit followed by Red ware and Grey ware at the site. The potteries are wheel thrown and medium to fine in fabric, semi-oxidized to oxidized. Important shapes include vase, dish, *handi* etc. in few instances black slipped and black polished ware noticed during the spade work.



Fig.16. Decorated potsherds

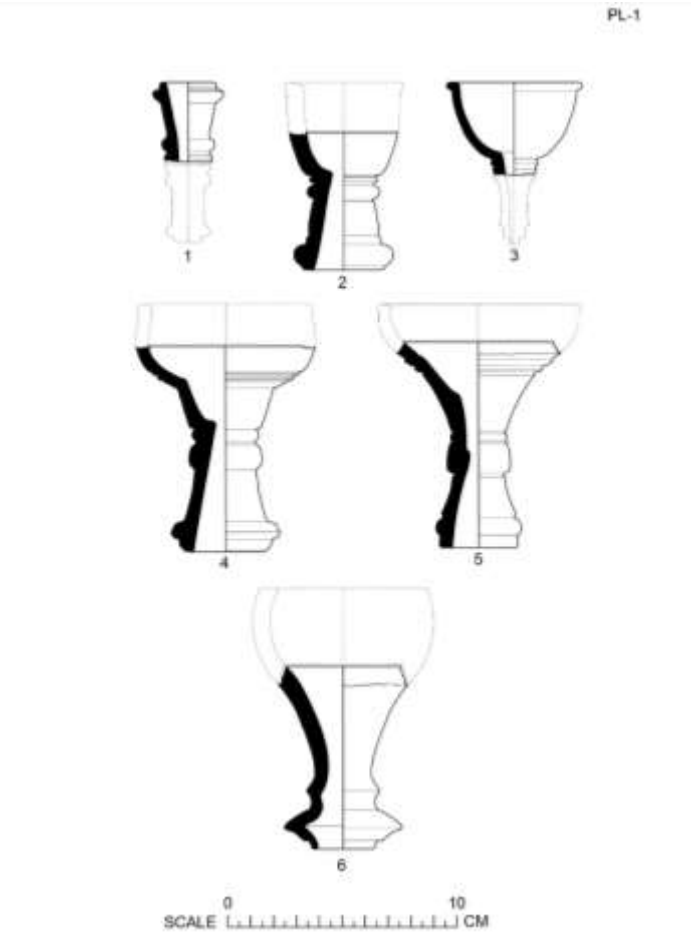


Fig.17. Terracotta Smoking pipe

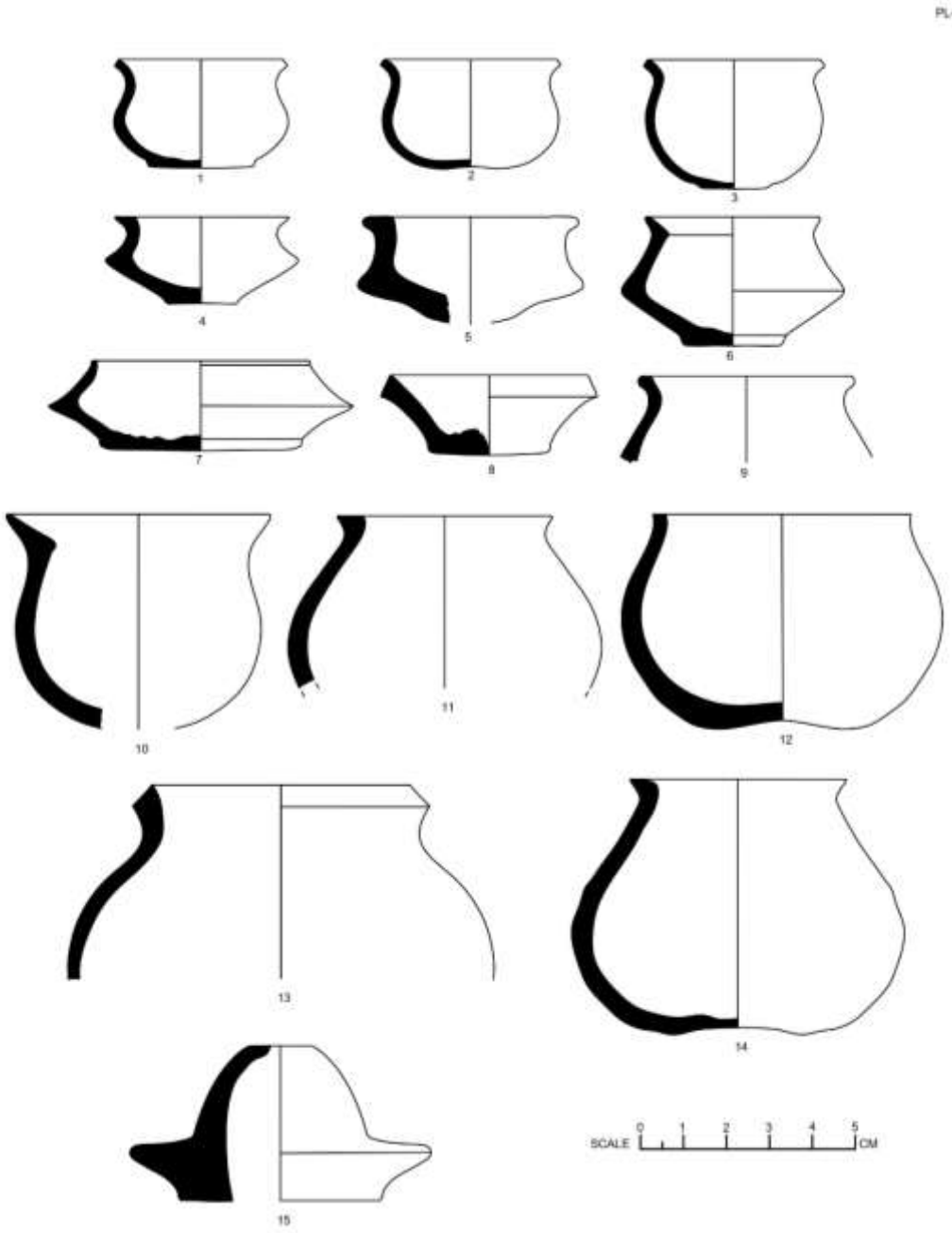


Fig.18. Miniature Pot: Sl. 1- 3, 10 Grey Ware; Sl. 4- 9, 11- 15 Red ware

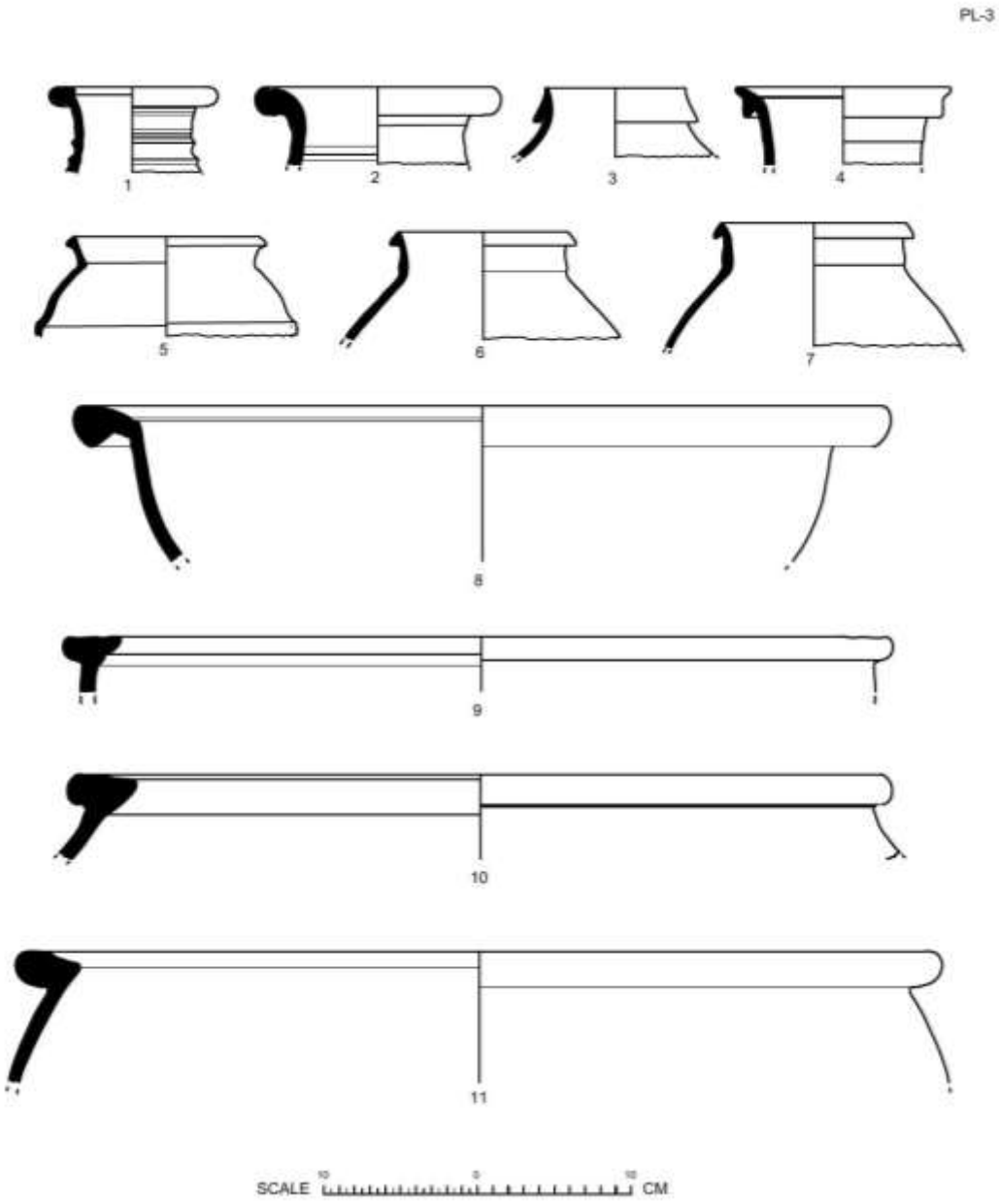


Fig.19. Pottery: Sl. 1- 3, 8-11 Grey Ware; Sl. 4- 7 Red ware

PL-4

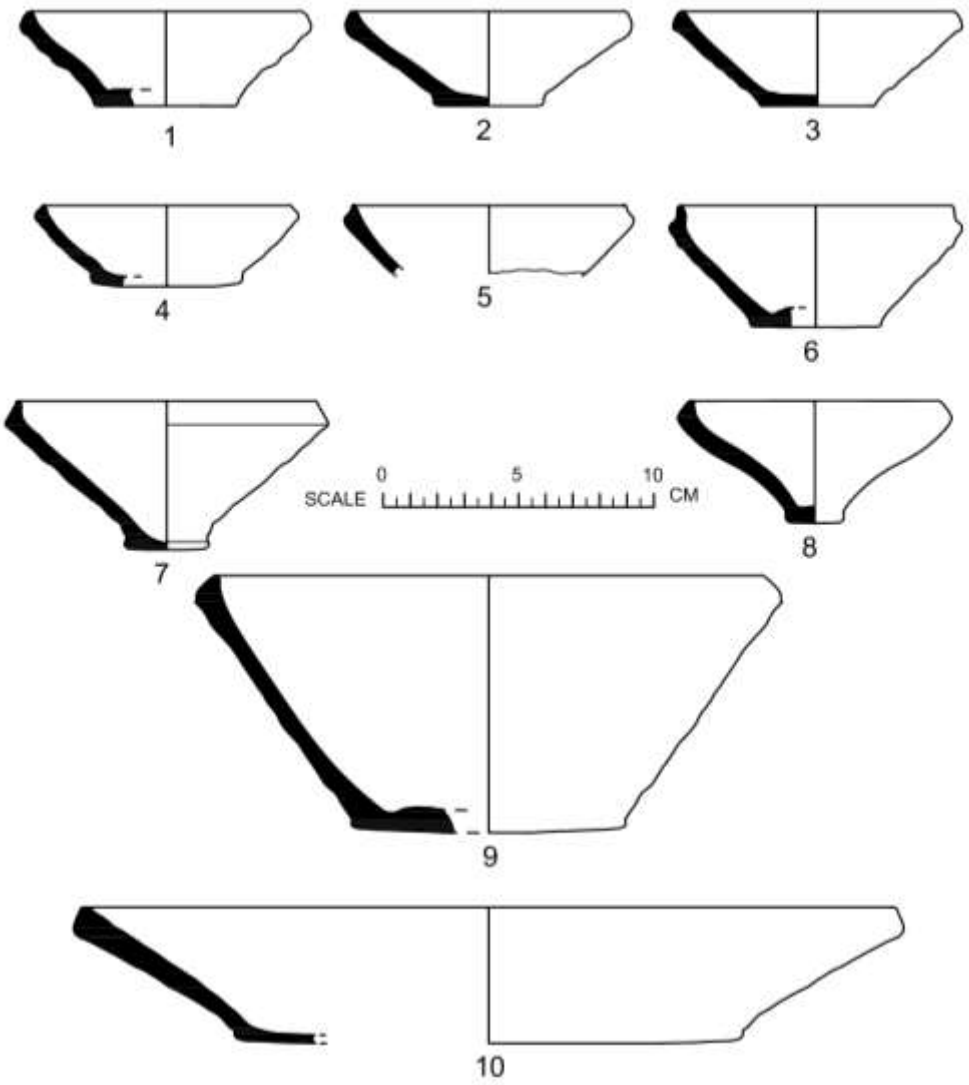


Fig.20. Pottery: Sl. 1- 7 Red ware; Sl. 8- 10 Grey Ware

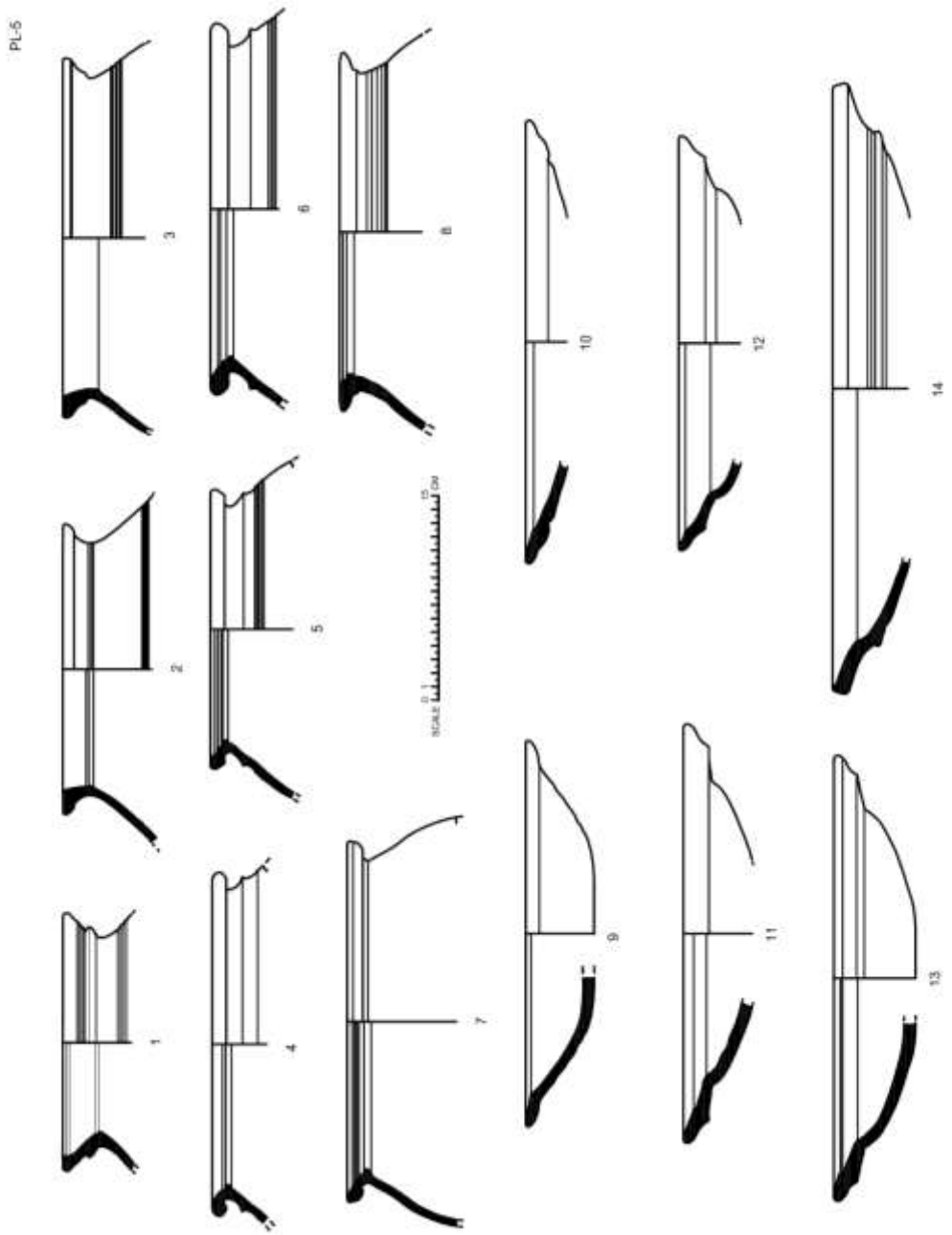


Fig.21. Pottery: Sl. 1- 3 Grey Ware; Sl. 4- 14 Red ware

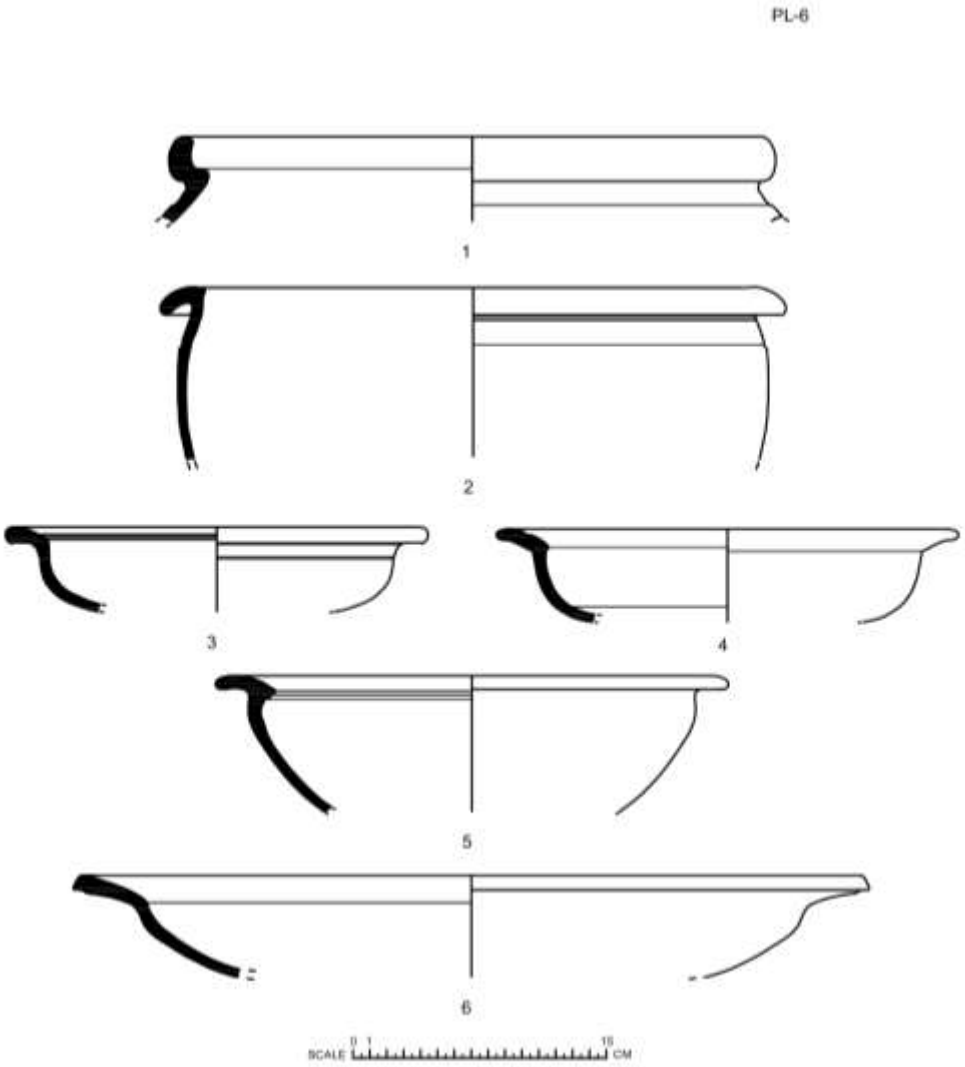


Fig.22. Pottery**:** Sl. 1- 6 Grey Ware

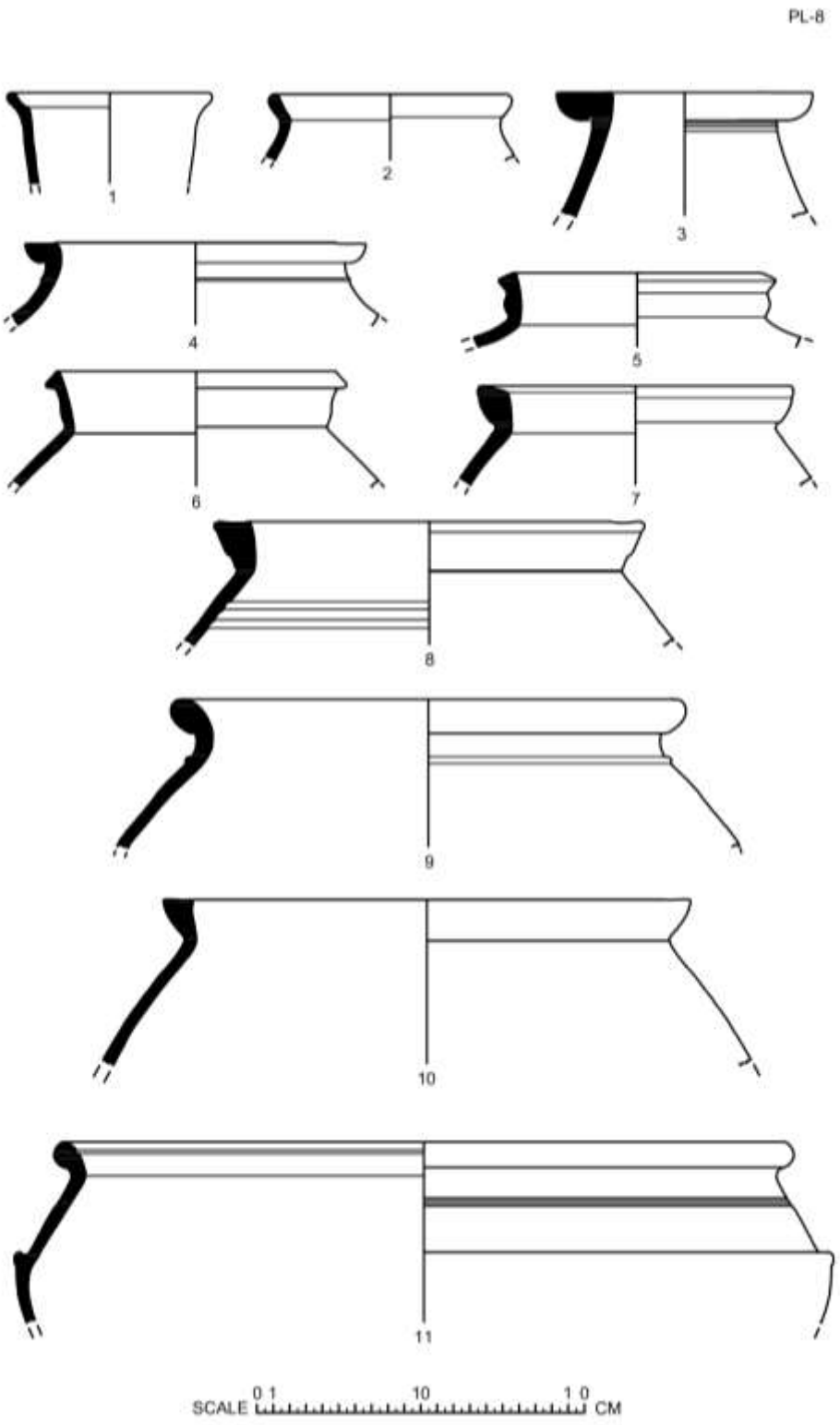


Fig.23. Pottery**:** Sl. 1-11 Red ware

Stone Ware

Excavation revealed few sherds of stone ware. Soft soap stones are used as raw materials and they are prepared by chiselling, grinding and polishing method. The sherds have lustrous polish in the inner and outer surface. Similar types of stone ware found from Ratnagiri, Lalitgiri, Chudangagarh, Khalkatpatna. One stone ware dish was found during the excavation at Barabati has traces of orange colour painting in the outer body of the ware.



Fig.24. Stone Ware

Terracotta Lamps

In medieval period, lamps made out of clay were generally resembles like miniature bowls with concave mouth for wicks and some have decorated with incised and applique patterns. Few lamps have soot mark at the tip. One of the dominant types of lamps found at the site is carinated in the centre having wall like features of the rings, flat base found in the earlier deposit while the lamps found in the upper cultural level have concave (blind hole) at the base which may be used for holding the lamps in the finger or in stands. Such type of lamps mostly found in the mediaeval sites of northern India. The dull red coloured lamps are outnumber in comparison to dark grey lamps at the site throughout the cultural deposit.



Fig.25. Terracotta Lamps

Chinese Ceramics

One of the significant findings during the excavation is Chinese ceramics, both painted and plain varieties of porcelain and celadon. The colours of this ceramics varies from whitish to bluish white; important shapes include dish, shallow dish, bowl, cuplets and are decorated with floral, structural (pagoda), cross, scenic paintings in the inner as well as outer surface of the ceramics. Few inscribed porcelain resembles with ancient Chinese hanzi/kanji (script). Most of the coastal medieval sites of Odisha revealed celadon sherds such as at Manikapatana, Khalkatpatna in the district of Puri. Few sherds recovered from the sites have exhibited poor workmanship both in case of raw material and textures which implies they might be arrived via maritime contact with other Southeast Asian countries. Celadon found at Barabati excavation have resembled with the Manikapatana which are dated to the 13th and 14th century, corresponds with the Yuvan and southern Song dynasties (1127–1279). It is believed by few historians that these ceramics found from different hinterland sites and coastal settlements due to strong trade interlinks as well as it was best gift goods from early mediaeval to late mediaeval period by the traders to kings and nobles. These ceramics generally used by noble class people during that time (K. K. Basa and K. S. Behera, 2000. Maritime Archaeology of Odisha, published in K. K. Basa and P. Mohanty (eds) Archaeology of Orissa Vol. 1 Pp. 573-574). In layer (6) best specimens of Tang white ware was retrieved during excavation associated with pottery and material culture of

circa 7th to 9th century CE at the site. These Tang white ceramics were produced during 6th to 9th century CE in China and circulated outside of China. The evidence of such ceramics not only gives direct or indirect maritime contact of Odisha but also push back the antiquity of trade relationship with outside world. The material evidence and the religious environment of Odisha was more revolving around Buddhism. The visit of Hiuen-Tsang to Odisha in 6th Century CE proved that the religious landscape of Odisha, India was rich in Buddhism and Buddhist establishment. The Buddhist establishment i.e. Lalitagiri, Langudi, Udayagiri and Ratnagiri were well flourished sites during early historic period to mediaeval period while comparing the material culture retrieved from the early levels of Barabati fort, resembling with the material culture of above mentioned sites. The stable religious thought of the time based on Buddhism in Odisha may be one of the causes of trade relationship with South East Asian Countries and beyond. Discovery of Chinese pottery in Odishan Coast in early mediaeval period was due to well-connected maritime trade network between China, South-east Asian Countries and Persian Gulf and East African regions.

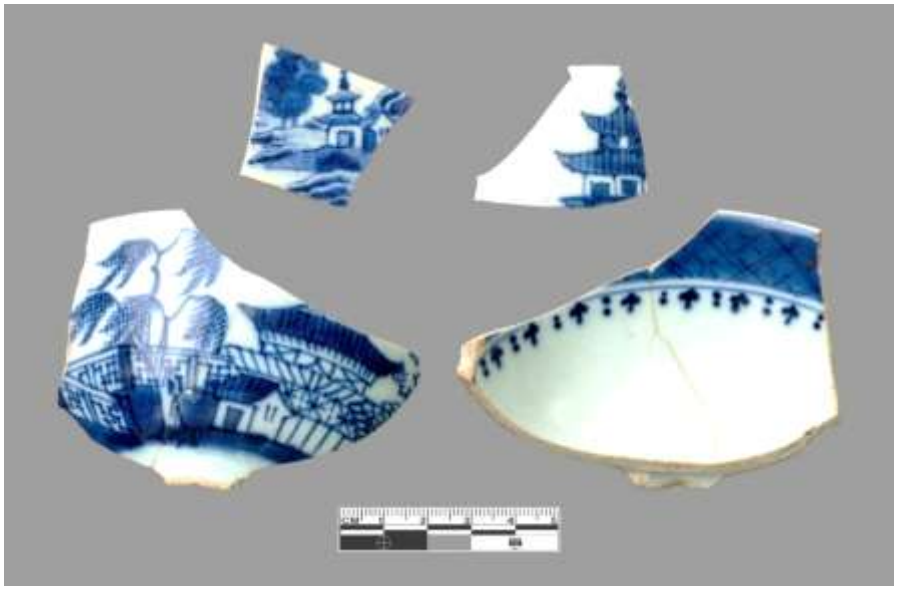


Fig. 26. Chinese porcelain



Fig. 27. Tang white ware

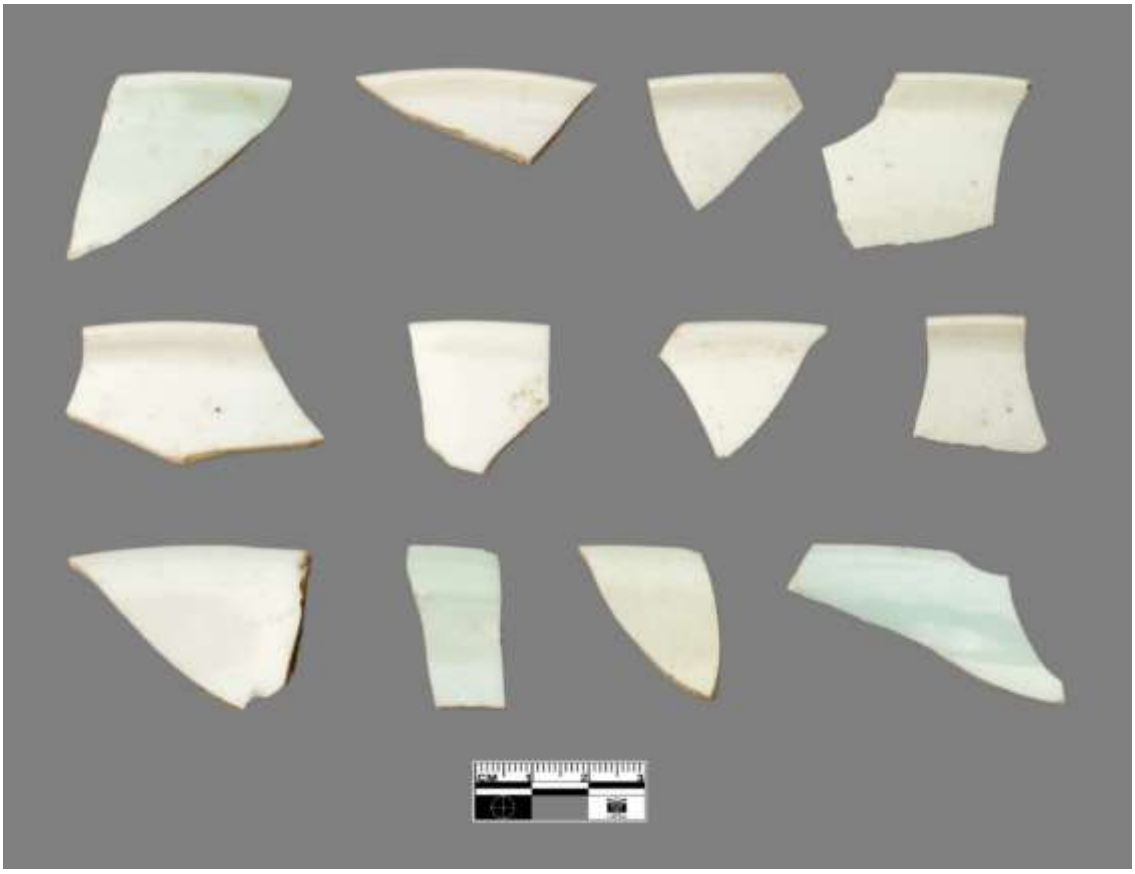
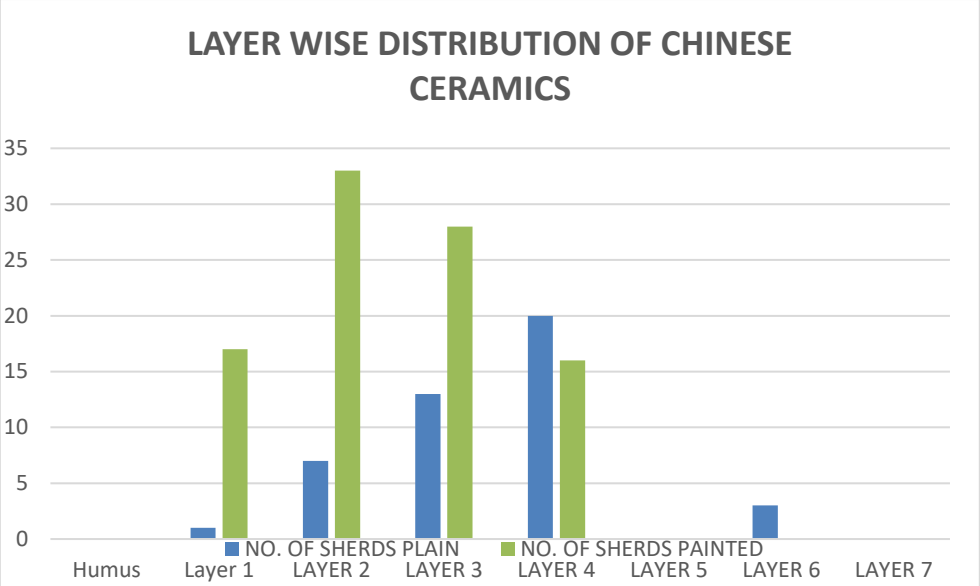


Fig.28. Plain Chinese ware



Fig.29. Burmese ware

LAYER	TOTAL NO. OF CHINESE SHERDS	
	PLAIN SHERDS	PAINTED SHERDS
Humus	0	0
Layer 1	1	17
Layer 2	07	33
Layer 3	13	28
Layer 4	20	16
Layer 5	0	0
Layer 6	3	0
Layer 7	0	0



Features of the site:

1. It is a well-fortified site of early medieval period having a moat around it.
2. This excavation pushes back the antiquity of the site to circa 7th century CE on the basis of material culture.
3. The present excavation revealed seven cultural layers without any hiatus starting from the top to bottom with 4.25m cultural deposit (below the present road level).
4. The evidence of ceramics of foreign origin shows Odisha had maritime contact with South East Asian Countries and beyond.
5. One of the significant findings during the excavation is inscribed Chinese ceramics as well as the best specimens of Tang white ware retrieved from the bottom layer which was produced during 6th to 9th century CE in China and circulated outside of China.
6. The architectural members, sculptural fragments unearthed from the site, suggests that more than one temple of different time period existed in the site.
7. The monumental architecture noticed in the sites are in the form of remains of temples, buildings, pathways, platforms, pillars made out of sand stone, laterite and khondolite stone.

8. The ceramic assemblage yielded from the excavation are of red ware, black ware, black slipped ware, grey ware, variant potteries of porcelain, celadon, Burmese ware, white ceramics and glazed ware of overseas origin.

Conclusion

The strategically located settlement has its beginning from circa 7th century CE and grown up to a well-fortified medieval settlement during the time of Gangas. This fort once formed the seat of authority under the mighty Gangas known as Abhinava Varanasi Kataka, as mentioned in the copper plate grant of Anangabhimadeva – III. It witnessed the fortune and fall of the dynasties such as Bhaumakaras, Somavamsis, Gangas, Suryavamsis, Gajapatis, Bhois, Afghans, Mughals, Marathas and finally the British till the beginning of 20th century CE. During the British Raj who made the fort as a prison of confinement of several illustrious rulers of the land. The Raja of Kujanga in 1800, the Raja of Khurda (Ramachandra Deva - II) in 1818 and the Raja of Surguja in 1819 with his family members were kept under strict confinement. Prior to this the fort also faced the onslaught of Firoz Shah Tughluq in 1361 CE during the Ganga ruler Bhanudeva-III as mentioned in the Tarikh- i- Firuz Shahi written by Shams-i-Siraj-Afif. Mukunda Harichandan of Bhoi dynasty (1560-1568 CE), the last Hindu king had profusely added to the buildings and the fortifications of the Barabati. This statement was corroborated with the accounts given in Ain-i-Akbari by Abul Fazal Allami (translated by Colonel H.S. Jarrett) Vol II that Raja Mukundadeva built a palace having nine segments, first one for elephant stable, the second one occupied by artillery, guards and attendants and the third one for patrol and gatekeepers, the fourth one for workshops, the fifth one for kitchen, the sixth one for public reception, the seventh one for private apartment, the eighth one for women apartment and the ninth one for sleeping chamber of governor/king.

The imprint of the glorious monumental architecture of Barabati fort blow down by the Colonial bureaucrats. The destruction of the fort was intensified in the early phase of British rule. In 1828 under the suggestions of Peckenham, the Commissioner, the stones of the fort wall and structures were utilized for the construction of the public roads and even sold to the public at a price of 5 to 6 rupees per hundred pieces. In 1829, the Light House at the False point and the Cantonment road of Cuttack were constructed with the fallen stones of the fort. In 1837, the Superintendent of Embankments caused further destruction by removal of the remains of the walls of the fort. In 1940's the State Public Works Department constructed staff quarters and workshops as well as the district administration built Cuttack Club in the northern part

of the fort. Later the entire southern part of the fort has been occupied by Satyabrata and Jawaharlal Nehru Indoor Stadium.

Barabati fort faced many challenges from external invasions, natural calamity like flood and urban pressure at present. Each structural edifices stands in mute providing information of its rise and fall. The material remains unearthed during excavation highlights the maritime glory of ancient Odisha (Odra- Kalinga). The excavation in Area II, revealed seven cultural layers without any hiatus starting from the top to bottom with 4.25m cultural deposit. On the basis of ceramic and structural remains the whole cultural deposit can be assigned to circa 7th century CE to early 20th century CE. One of the significant findings during the excavation is inscribed Chinese ceramics as well as the best specimen of Tang white ware retrieved from the bottom layer was produced during 6th to 9th century CE in China and circulated outside of China. The evidence of such ceramics not only gives direct or indirect maritime contacts of Odisha but also push back the antiquity of trade relationship with outside world. The Buddhist establishment of Lalitagiri, Langudi, Udayagiri and Ratnagiri were well flourished sites during early medieval period while comprising the material culture retrieved from the early levels of Barabati fort, it resembles with the material culture of above mentioned sites. The stable religious thought of the time based on Buddhism in Odisha may be one of the factors of trade relationship with South East Asian Countries and beyond. The evidence of Chinese pottery in Odishan Coastal sites and the depiction of giraffe carved on the wall of Sun Temple Konark built in 13th century suggested well-connected maritime trade network between China, South-east Asian Countries, Persian Gulf and East African regions. The excavated remains at Barabati Fort reveal to the fact that most probably the tradition of Boita Bandana, Bali Yatra started from this area in particular and from the coastal eastern India in general.

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References

Behera, Karuna Sagar and Jagannath Patnaik, Harish Chandra Das. 1990. *Cuttack One Thousand Years, Vol. II.* Cuttack: Cuttack Millenium Celebrations Committee.

Behuria, N. C. 1996. *Orissa District Gazetteer, Cuttack.* Cuttack: Gazetteers Unit, Department of Revenue, Government of Orissa.

Chhabra, B. C. and D C Sircar, 1949-50. Nagari Plates of Anangabhima III, Saka 1151 and 1152, *Epigraphia Indica* (EI) Vol. XXVIII, pp. 235-258

Dash, R. N. 2017. November. Baliyatra: A Festival of Odisha's Ancient Maritime Trade. *Odisha Review*, pp. 13-14.

Elliot, H. M. 1871. *History of India, Vol. III.* London: Trübner and Co.

Garnayak, D. B. 2022. Excavation at Ulta- Kheda, Hastinapur: 2021. *Puratattva, No. 52*.

Indian Archaeology: 1989-90: 1990-91, 1991-92, 1992-93, 1993-94, 1994-95,1995-96, 1996-97, 2003-04.2006-07- A Review

Jarrett, Colonel H.S. 1891. The Ain-I-Akbari Vol. II by Abul Fazal Allami (translated from the original persian), published by the Asiatic Society of Bengal, Calcutta.

Majhee, B. 2021. A Study of Geographical Significance and Evolution of the City of Cuttack. *International Journal Humanities and Social Science Invention*, pp.1- 4.

Mehta, R. 1979. *Mediaeval Arcaheology.* Delhi: Ajanta Publications.

Mishra,P.K. and J.K. Samal, 1997. *Comprehensive History and Culture of Orissa Vol. I; Part II (Early Times to 1568).* New Delhi: Kaveri Books.

Patnaik, J. K. 2016. *Excavations at Lalitgiri (1985-1991).* New Delhi: The Director General, Archaeological Survey of India.

Rao, K.V.B. 2000. Excavation at Barabati Fort, Cuttack, published in K. K. Basa and P Mohanty (eds), *Archaeology of Orissa*, Delhi: Pratibha Prakashan, pp. 664,

Sinha, B.N. 1990. Geography of Cuttack City, pub. in *Cuttack one Thousand years* Vol. -I, pp. 1-4

Valenstein, S. G. (1975). *A Handbook of Chinese Ceramics.* New York: Samuel I. Newhouse Foundation.

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